SAMPLE — SYNTHETIC FIELD DATA FOR DEMONSTRATION · NOT A WETLAND DELINEATION · NOT FOR REGULATORY SUBMITTAL

WETLAND DELINEATION — SAMPLE MAP SET (MINNESOTA)

Sample site · Chisago County, Minnesota · USACE St. Paul District / MN BWSR joint guidance figure set

SAMPLE — synthetic field data for demonstration; not a wetland delineation; not for regulatory submittal. Wetland boundaries, flags, sample points and photo points were generated from public NWI, SSURGO and 3DEP data to show map-set formatting only. No field survey was performed, no scientist certified this work, and no jurisdictional status is indicated. The base location lies inside the Carlos Avery Wildlife Management Area boundary (Sunrise Unit, MN DNR); no finding is made about the property.

PROJECT INFORMATION

Item	Value
Site	Sample site — demonstration only (no owner, parcel or address data used)
Land status	39.83 of the 39.85 ac lie inside the Carlos Avery Wildlife Management Area boundary (Sunrise Unit, MN DNR; layer bdry_dnr_wildlife_mgmt_areas_pub, MN Geospatial Commons, edited 2024-01-25, queried 2026-10-09); PAD-US 4.1 records MN DNR fee ownership on 36.17 ac (2015 boundary). Field data synthetic; no finding about the property.
County, State	Chisago County, Minnesota
USACE district	St. Paul District (MVP), Mississippi Valley Division
Review area	NE¼ NW¼ Sec. 14, T34N R21W, Extended Fourth P.M. (BLM aliquot part); 39.85 acres
Approx. centre	45.437977° N, 92.930375° W (NAD83(2011))
USGS 7.5′ quadrangle	Stacy, MN (45092-D8)
Wetlands (synthetic)	5 wetlands, 14.0000 ac (Circular 39 types 1, 2, 6, 7; Eggers & Reed communities listed on Sheet 8)
Horizontal datum	NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database)
Vertical datum	NAVD88, feet (US survey)
Imagery	USDA NAIP 2023-07-11, 0.3 m (recent); 1991 DOQ, NAIP 2010-09-12 and 2021-06-18 (historical)
Elevation	USGS 3DEP 1-m DEM; lidar work unit MN_CentralMissRiver_1_B22, flown 2022-05-14 to 2022-05-27
Map standard	BWSR + USACE St. Paul District joint guidance v2.0 (2015) figure list; USACE HQ Recommended Minimum Standards for ARDRs (July 2025) add-on items
Prepared by	Danir Kantserov — GIS/CAD mapping · Independent contractor (white-label sample) · prepared 2026-10-09 · Rev. 3.1 (2026-10-10)

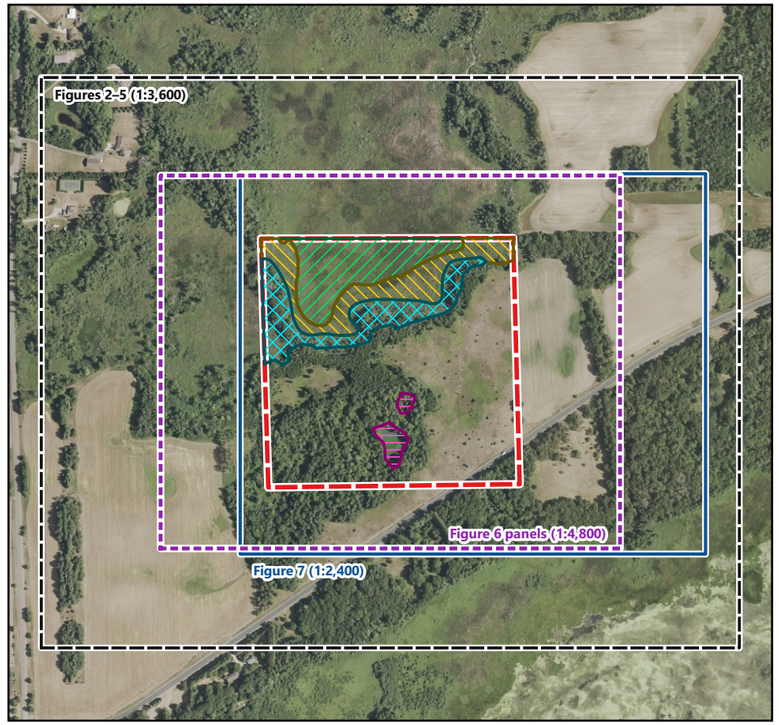
SHEET INDEX

Sheet	Title	Scale
1	Cover sheet, project information and sheet index	—
2	Figure 1 — Site location with directions, S/T/R and corner coordinates (USGS Stacy quadrangle)	1 in = 2,000 ft
3	Figure 2 — Topography: lidar contours and shaded relief	1 in = 300 ft
4	Figure 3 — NRCS soils with hydric rating and soil-unit table	1 in = 300 ft
5	Figure 4 — National Wetlands Inventory and MN Public Waters Inventory	1 in = 300 ft
6	Figure 5 — FEMA flood hazard (DFIRM) context and FIRM panels	1 in = 300 ft
7	Figure 6 — Recent and historical aerial review (1991, 2010, 2021, 2023)	1 in = 400 ft
8	Figure 7 — Final delineation on 2023 NAIP, sample points keyed to data forms	1 in = 200 ft

DIGITAL DELIVERABLES

File	Contents
SAMPLE_MN_Delineation_MapSet.pdf	This 8-sheet set, 11 × 17 in, vector PDF; sheets\ holds the same pages one by one. Rev. 3.1 (2026-10-10) on every title block.
gis\MVP-SAMPLE_<layer>_20261009.shp	13 ESRI shapefile layers in EPSG:6501 (US survey feet)
gis\MVP-SAMPLE_Delineation_WGS84_20261009.gpkg	The same layers in WGS 84 (G2139), one GeoPackage
gis\MVP-SAMPLE_metadata_20261009.txt	Datum, projection, sources, classification notes and mapper contact; Rev. 3.1 (2026-10-10)
Aquatic_Resources_Table.csv	The 11 columns of the ORM bulk-upload sheet (WGS 84 (G2139), 6 decimals)
SAMPLE_MN_Delineation.dxf	Drawing in AutoCAD 2013 DXF format, US survey feet, named layers; Rev. 3.1 (2026-10-10) in its note. Opens without errors in the ODA engine; not verified in AutoCAD.
Checklist and QA records	Requirement checklist, number reconciliation and CAD checks: available on request

KEY MAP — SHEET COVERAGE (1 inch = 800 feet)



TRUE NORTH

0 500 1,000 1,500 2,000 ft

WETLAND SUMMARY (Sheet 8)

Wetland	Community	Cowardin	Acres
Wetland 1	Sedge meadow	PEM1B	4.8411
Wetland 2	Shrub-carr	PSS1C	4.1639
Wetland 3	Hardwood swamp	PFO1B	4.2831
Wetland 4	Seasonally flooded basin	PEM1C	0.1630
Wetland 5	Seasonally flooded basin	PEM1C	0.5489
Total			**14.0000**

REVISIONS

Rev.	Date	Change
3	2026-10-09	Issued as the sample set (revisions 0–2 of the same day were internal drafts and were not issued)
3.1	2026-10-10	Land status worded as its sources state; WMA source listed; WMA band on Figure 7

QUALITY CHECKS (automated, run on every build). Areas, lengths, coordinates and table rows are recomputed from the delivered GIS files in two independent passes; wetland types are checked against the BWSR crosswalk (2025); SHP, GeoPackage and DXF geometry is tested (valid, no overlaps, flags on their wetland edge); labels and boxes are tested on the printed sheets (each label on its own feature, leaders reaching their text, no label over a point symbol, flag, arrow or boundary line, no overlaps, no box over the review area, a delineated feature, a label or a control point); the historical images are checked against road centrelines; every character is on an allowlist; the DXF opens without errors in the ODA engine (DXF to DWG to DXF round trip unchanged; not verified in AutoCAD). Each check added after a review is also run on the earlier revision and must catch the defect it was written for.

WETLAND DELINEATION MAP SET — MINNESOTA

Sample site · Chisago County, Minnesota · NE¼ NW¼ Sec. 14, T34N R21W

Sample site — demonstration only (SYNTHETIC field data)

USACE St. Paul District + MN BWSR · joint delineation report guidance v2.0 (2015)

COVER SHEET
PROJECT INFORMATION AND SHEET INDEX

LEGEND

- Review area (sample site, 39.85 ac)
- Wetland 1 — Sedge meadow, PEM1B (4.8411 ac)
- Wetland 2 — Shrub-carr, PSS1C (4.1639 ac)
- Wetland 3 — Hardwood swamp, PFO1B (4.2831 ac)
- Wetland 4 — Seasonally flooded basin, PEM1C (0.1630 ac)
- Wetland 5 — Seasonally flooded basin, PEM1C (0.5489 ac)

Coverage of the other figures

- Figures 2–5 (1:3,600)
- Figure 7 (1:2,400)
- Figure 6 panels (1:4,800)

DATA SOURCES (all public; see the notes on each figure)

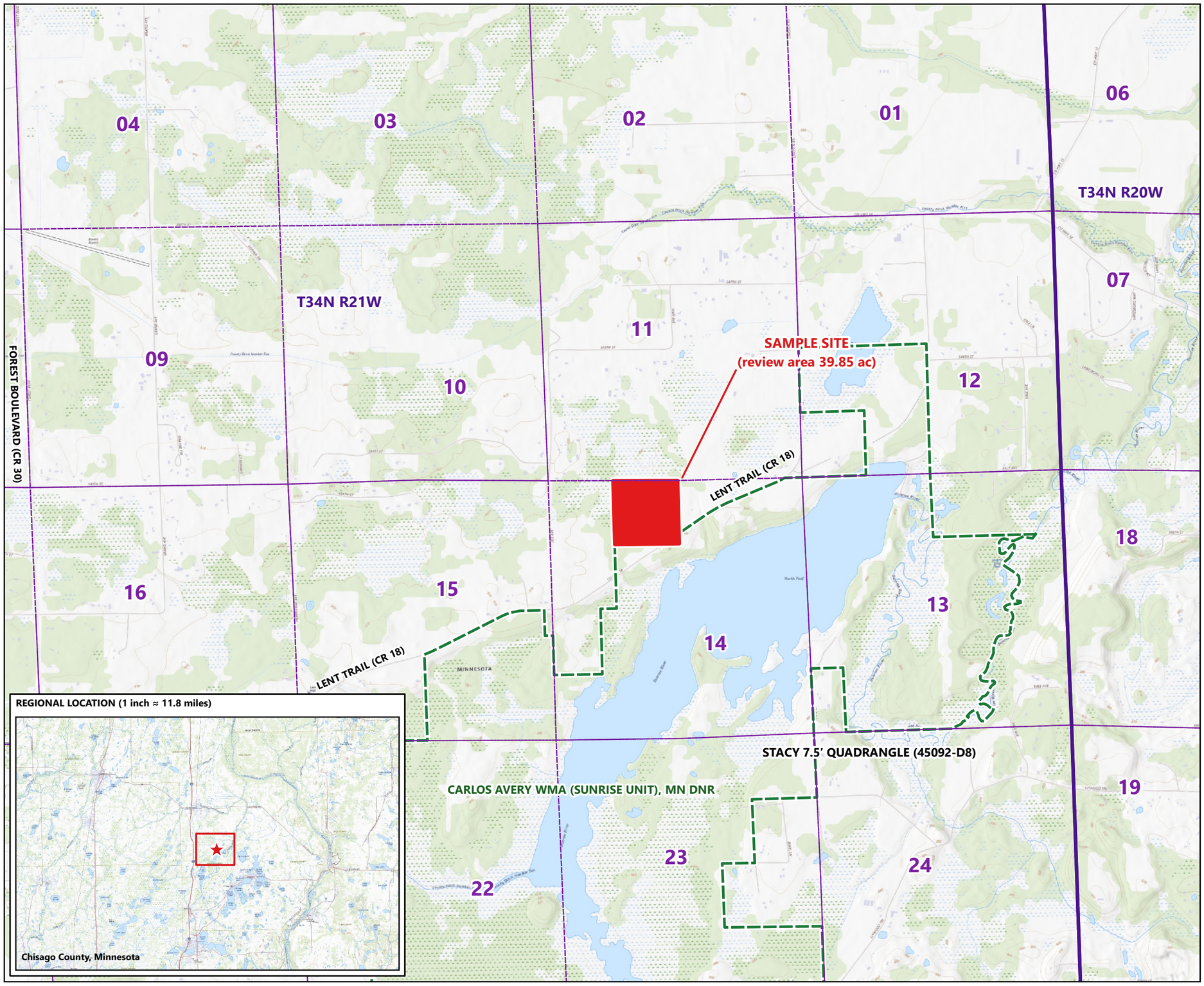
Layer	Source and date
Aerial imagery	USDA NAIP 2023-07-11 (0.3 m), 2021-06-18, 2010-09-12 via the Planetary Computer STAC; 1991 USGS DOQ via the MnGeo WMS
Elevation	USGS 3DEP 1-m DEM, lidar work unit MN_CentralMissRiver_1_B22, flown 2022-05-14 to 2022-05-27
Soils	USDA NRCS SSURGO MN025, version 23 established 2026-09-07
Wetland inventory	USFWS NWI, project "MN Update" (2010 CIR imagery), queried 2026-10-09
Public waters	MN DNR Public Waters Inventory (MnGeo), queried 2026-10-09
Flood hazard	MN DNR copy of the FEMA DFIRM 27025C; panels 27025C0260D, 27025C0275D effective 2012-04-17
PLSS	BLM National PLSS (CadNSDI), queried 2026-10-09
Land status	MN DNR WMA boundaries (MN Geospatial Commons), queried 2026-10-09; USGS PAD-US 4.1, fee layer (2015 boundary)
Roads, places	OpenStreetMap contributors (ODbL), extract 2026-10-09
Survey control	NGS mark DL2419 "CHICO 22", last recovered 2020-04-22
Field data	SYNTHETIC: boundaries, flags, sample and photo points generated 2026-10-09

This sample shows the desk (GIS/CAD) part of a Minnesota wetland delineation report: the figure set required by the BWSR / St. Paul District joint guidance, drawn from public data, plus the GIS, DXF and ORM files.
All field data in this set are synthetic. A real delineation requires field work under the 1987 Corps Manual and the Northcentral and Northeast Regional Supplement by a qualified wetland delineator.
Coordinates: NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database). Latitude/longitude: decimal degrees, NAD83(2011); the GeoPackage and the ORM CSV are in WGS 84 (G2139), transformed with PROJ at epoch 2026.77 (1.4 m from NAD83(2011) here).
Elevations: feet (US survey), NAVD88; USGS 3DEP 1-m DEM from lidar work unit MN_CentralMissRiver_1_B22 (QL 1), flown 2022-05-14 to 2022-05-27; source DEM published 2023-05-18 (USGS 3DEP index, WESM).

Prepared and revised by: Danir Kantserov — GIS/CAD mapping
Organization: Independent contractor (white-label sample)
Date prepared: 2026-10-09 · Rev. 3.1 (2026-10-10)
Field data: SYNTHETIC (no field survey) · Scientist of record: none
Map file: SAMPLE_MN_Delineation_MapSet.pdf

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WETLAND DELINEATION MAP SET — MINNESOTA

Sample site · Chisago County, Minnesota · NE¼ NW¼ Sec. 14, T34N R21W

Sample site — demonstration only (SYNTHETIC field data)

USACE St. Paul District + MN BWSR · joint delineation report guidance v2.0 (2015)

SITE LOCATION MAP WITH DIRECTIONS (FIGURE 1)

LEGEND

- Review area / sample site (39.85 ac)
- Carlos Avery WMA boundary (Sunrise Unit), MN DNR
- PLSS section line (BLM)
- PLSS township boundary (BLM)
- Site location (regional inset)
- Extent of main map

SITE LOCATION
Section 14, T34N R21W (Extended 4th P.M.), NE¼ NW¼
Chisago County, Minnesota
USGS Stacy 7.5′ quadrangle (45092-D8)
Centre: 45.437977° N, 92.930375° W (NAD83(2011))

DIRECTIONS TO THE SITE (from Stacy City Hall, Stacy, MN)
1. Go north on Forest Boulevard (CR 30) for about 1.4 mi.
2. Turn right (east) onto Lent Trail (CR 18) and go about 2.9 mi.
3. The review area adjoins Lent Trail on the left (north-west side); it has no buildings.
Total about 4.3 mi. Routing: shortest path on the OpenStreetMap road network (extract 2026-10-09); verify before field use.

REVIEW-AREA CORNERS (NAD83(2011); EPSG:6501 FT)

Corner	Latitude (N)	Longitude (W)	Northing (ft)	Easting (ft)
NW	45.439793	92.932991	491,268.97	2,962,737.60
NE	45.439741	92.927806	491,272.08	2,964,068.73
SE	45.436178	92.927757	489,973.11	2,964,102.85
SW	45.436192	92.932926	489,956.12	2,962,776.13

Longitudes are west of Greenwich; the GIS and CSV files store them as negative numbers.

TRUE NORTH

0 1,000 2,000 3,000 4,000 ft

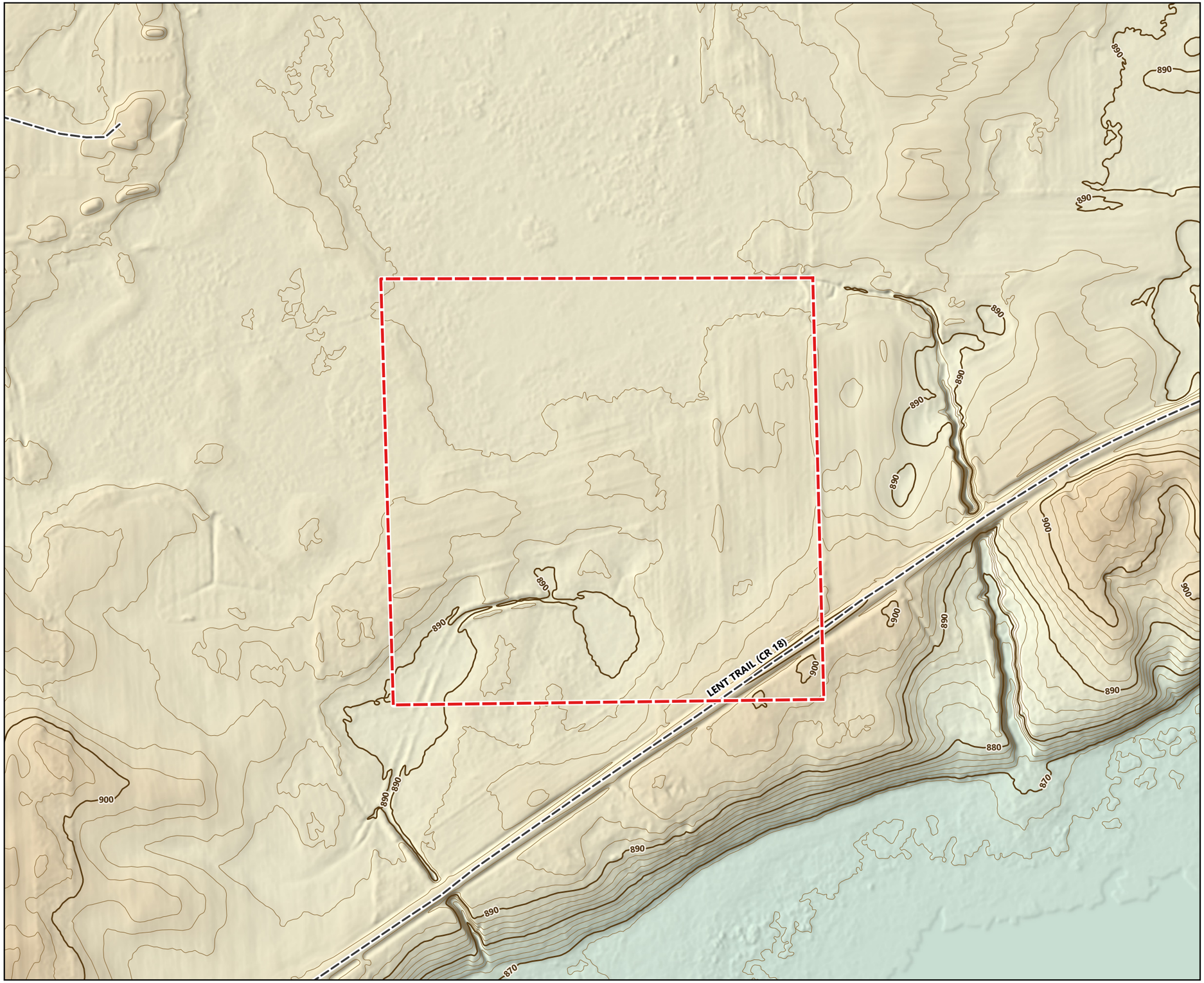
Scale: 1 inch = 2,000 feet (1:24,000)
when printed at 11 × 17 in (full size)

Location: Section 14, T34N R21W, Extended Fourth Principal Meridian (BLM PLSSID MN460340N0210W0), NE¼ of NW¼ (40-acre aliquot part from the BLM CadNSDI PLSS layer, queried 2026-10-09); county: Chisago, Minnesota; USGS 7.5′ quadrangle: Stacy (45092-D8). The review area is the aliquot polygon, not a legal survey.
Base map: USGS The National Map "USGS Topo" service, rendered 2026-10-09 for 1:24,000 (a US Topo-style base, not a scan of the printed quadrangle). Roads: OpenStreetMap contributors (ODbL).
Land status: 39.83 of the 39.85 ac review area lie inside the Carlos Avery Wildlife Management Area boundary (Sunrise Unit, MN DNR; layer bdry_dnr_wildlife_mgmt_areas_pub, MN Geospatial Commons, edited 2024-01-25, queried 2026-10-09); PAD-US 4.1 records MN DNR fee ownership on 36.17 ac (2015 boundary). The field data are synthetic; no finding is made about the property.
Coordinates: NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database). Latitude/longitude: decimal degrees, NAD83(2011); the GeoPackage and the ORM CSV are in WGS 84 (G2139), transformed with PROJ at epoch 2026.77 (1.4 m from NAD83(2011) here).

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WETLAND DELINEATION MAP SET — MINNESOTA

Sample site · Chisago County, Minnesota · NE¼ NW¼ Sec. 14, T34N R21W

Sample site — demonstration only (SYNTHETIC field data)

USACE St. Paul District + MN BWSR · joint delineation report guidance v2.0 (2015)

TOPOGRAPHY — LIDAR CONTOURS AND SHADED RELIEF (FIGURE 2)

LEGEND

- Review area (sample site, 39.85 ac)
- Contour, 2-ft interval (feet, NAVD88)
- Index contour, 10-ft interval, labelled (feet, NAVD88)
- Road (OpenStreetMap)

ELEVATION TINT (feet, NAVD88)

865 ft — 906 ft

Review area elevations (lidar DEM): 888.7 to 901.0 ft.
Relief: multidirectional hillshade of the 1-m DEM, vertical exaggeration x3, with the tint above (percentile 1–99 stretch of the 6,000 x 4,600 ft DEM window around the site).
Contours: 2-ft interval from the DEM smoothed by a 4-ft Gaussian; closed contours under 130 ft long are omitted. Lidar-derived contours are a mapping aid only; wetland boundaries are never taken from lidar alone (BWSR/MVP guidance, section 3.1).

LIDAR SOURCE (USGS 3DEP INDEX, WESM)

Item	Value
3DEP work unit	MN_CentralMissRiver_1_B22 (project MN_CentralMissRiver_B22)
Lidar collected	2022-05-14 to 2022-05-27, QL 1
Published	point cloud 2023-05-19; source DEM 2023-05-18
Vertical datum	NAVD88, GEOID18; shown in US survey feet
Contours	2-ft interval, 10-ft index, from the 1-m DEM
Lidar horizontal CRS	EPSG:6344, NAD83(2011) / UTM zone 15N
One-meter DEM specification	Meets (WESM); work units at the review-area centre: 1

TRUE NORTH

0 100 200 300 400 ft

Scale: 1 inch = 300 feet (1:3,600)

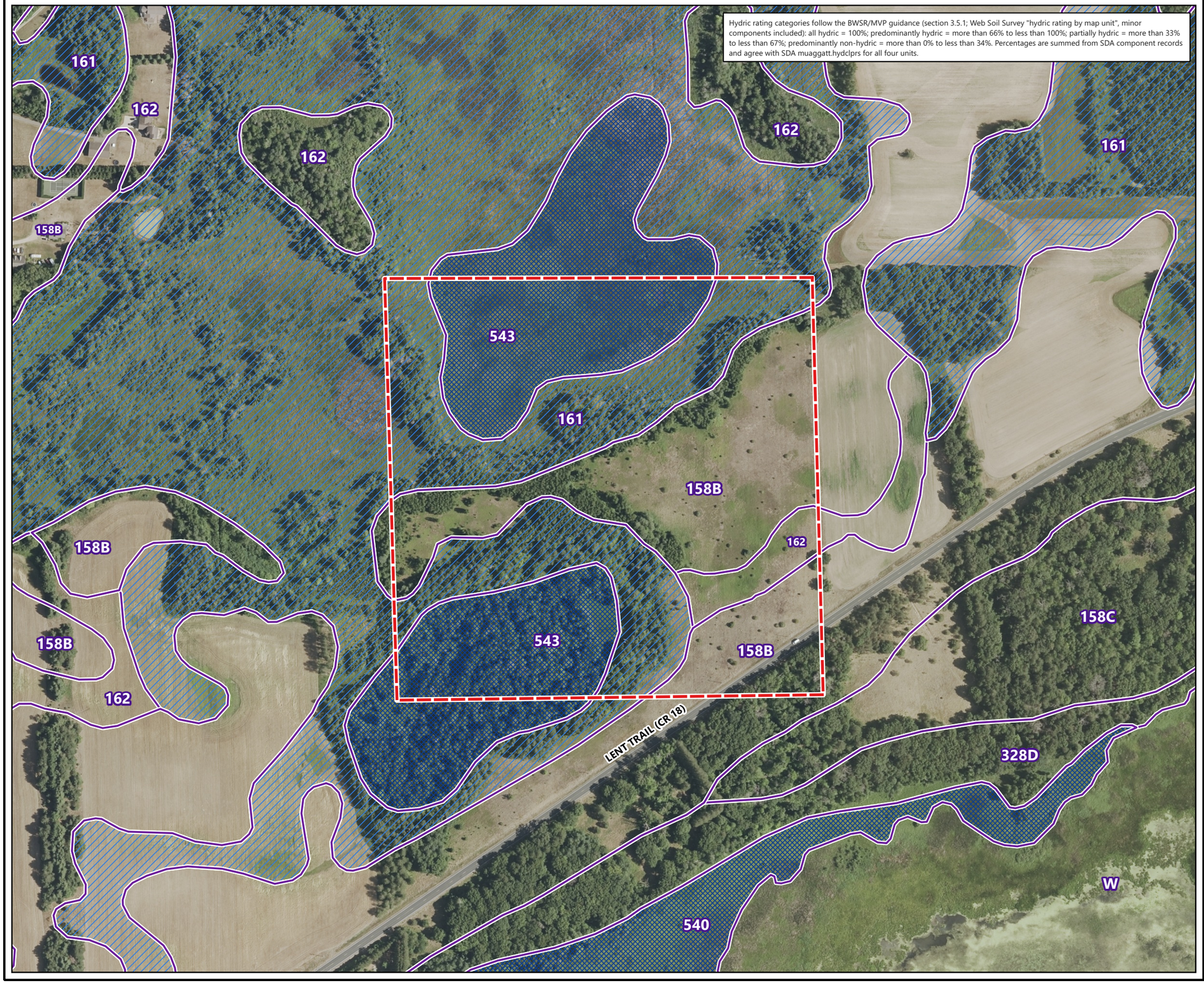
when printed at 11 × 17 in (full size)

Elevations: feet (US survey), NAVD88; USGS 3DEP 1-m DEM from lidar work unit MN_CentralMissRiver_1_B22 (QL 1), flown 2022-05-14 to 2022-05-27; source DEM published 2023-05-18 (USGS 3DEP index, WESM).
Topography is shown from LiDAR (this figure) and from the USGS Stacy 7.5′ quadrangle base (Figure 1, 1:24,000).
Coordinates: NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database). Latitude/longitude: decimal degrees, NAD83(2011); the GeoPackage and the ORM CSV are in WGS 84 (G2139), transformed with PROJ at epoch 2026.77 (1.4 m from NAD83(2011) here).
Roads: OpenStreetMap contributors (ODbL), extract 2026-10-09.

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WETLAND DELINEATION MAP SET — MINNESOTA

Sample site · Chisago County, Minnesota · NE¼ NW¼ Sec. 14, T34N R21W

Sample site — demonstration only (SYNTHETIC field data)

USACE St. Paul District + MN BWSR · joint delineation report guidance v2.0 (2015)

NRCS SOILS WITH HYDRIC RATING (FIGURE 3)

LEGEND

- Review area (sample site, 39.85 ac)
- All hydric
- Predominantly hydric
- Predominantly non-hydric

SOIL MAP UNITS IN THE REVIEW AREA (NRCS SSURGO, MN025)

Symbol	Map unit name	Acres	% of area	Hydric rating (% of components)	Components (% of unit)
158B	Zimmerman fine sand, 1 to 6 percent slopes	13.76	34.5	2% — Predominantly non-hydric	Zimmerman 90%; Cantlin 5%; Lino 2%; Isanti 2% (hydric); Soderville 1%
161	Isanti loamy fine sand	13.15	33.0	93% — Predominantly hydric	Isanti 90% (hydric); Soderville 4%; Lino 3%; Markey 3% (hydric)
543	Markey muck	11.54	29.0	100% — All hydric	Markey 90% (hydric); Warman 4% (hydric); Newson 3% (hydric); Isanti 3% (hydric)
162	Lino loamy fine sand	1.40	3.5	3% — Predominantly non-hydric	Lino 85%; Isanti 3% (hydric); Soderville 3%; Sartell 3%; Nymore 3%; Zimmerman 3%
Total		**39.85**	**100.0**		

SOILS SUMMARY

Hydric soils (units with more than 66% hydric components): 24.69 ac of 39.85 ac (62.0%)
Map unit symbols are labelled on the map (white on purple), each inside its own unit.
Categories not present in the review area: partially hydric, not hydric.

TRUE NORTH

0 100 200 300 400 ft

Scale: 1 inch = 300 feet (1:3,600)

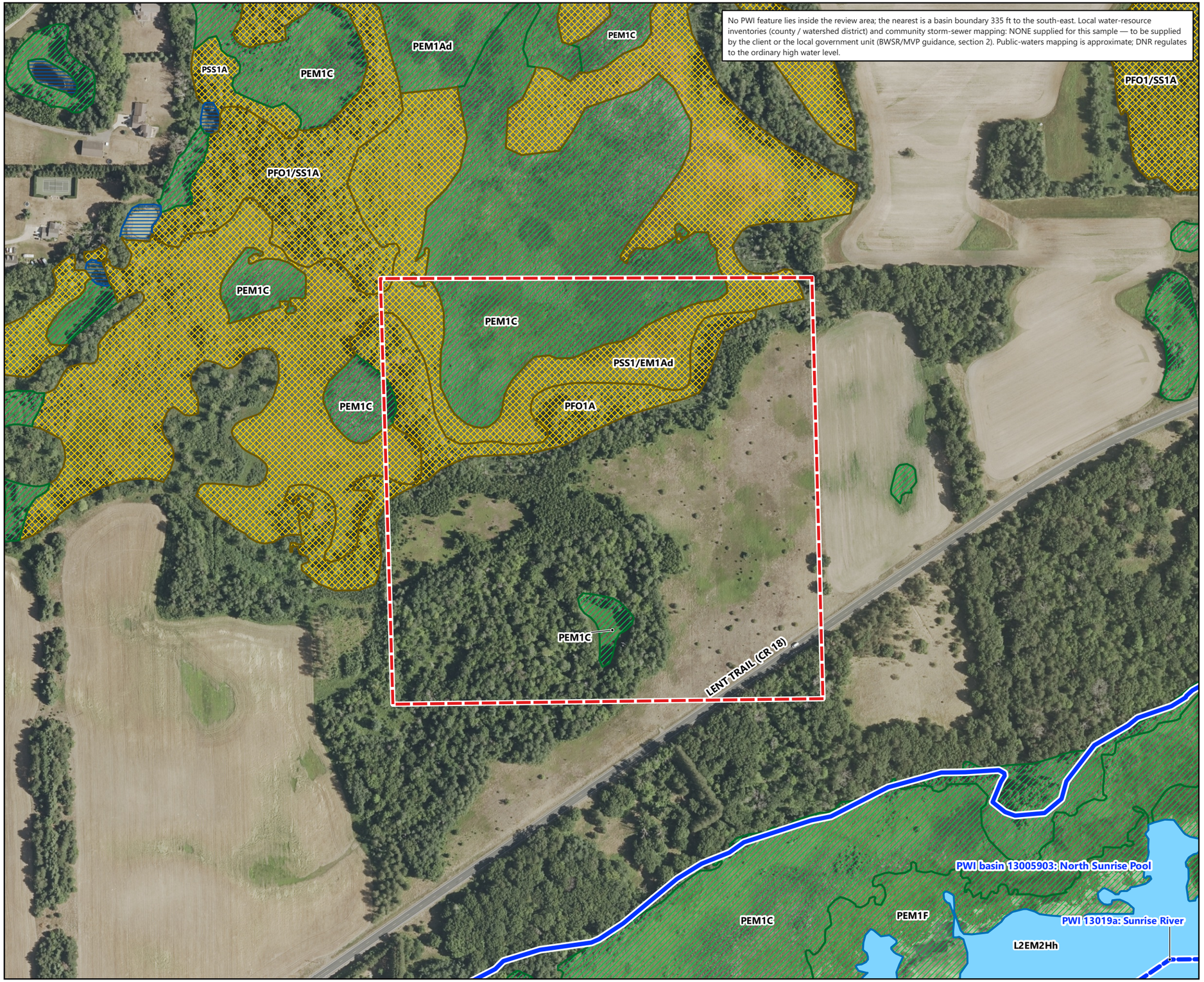
when printed at 11 × 17 in (full size)

Soils: USDA NRCS SSURGO, Chisago County, Minnesota (MN025), survey area version 23, version established 2026-09-07 (Soil Data Access, queried 2026-10-09). Map-unit polygons from the SDA WFS (mapunitpoly). A soil map shows where hydric soils may occur; it does not show wetland boundaries.
Imagery: USDA NAIP 2023, natural color, 0.3 m, orthorectified; acquired 2023-07-11 (quarter-quads m_4509233 NE, NW, SE, SW). Source: USDA-FSA-APFO tiles via the Microsoft Planetary Computer STAC (public). Mid-growing-season, leaf-on; no wet-season or leaf-off image was used.
Coordinates: NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database). Latitude/longitude: decimal degrees, NAD83(2011); the GeoPackage and the ORM CSV are in WGS 84 (G2139), transformed with PROJ at epoch 2026.77 (1.4 m from NAD83(2011) here).
Areas: planar, from the GIS geometry (EPSG:6501); acres rounded once (2 decimals for soils); totals are sums of the rounded rows shown.

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USACE St. Paul District + MN BWSR · joint delineation report guidance v2.0 (2015)

NATIONAL WETLANDS INVENTORY AND MN PUBLIC WATERS INVENTORY (FIGURE 4)

LEGEND

- Review area (sample site, 39.85 ac)
- Freshwater emergent wetland (PEM)
- Freshwater forested / shrub wetland (PFO, PSS)
- Freshwater pond (PUB, PAB)
- Lake / riverine (L, R)
- MN DNR Public Waters Inventory basin (outline)
- MN DNR Public Waters Inventory watercourse

NWI POLYGONS IN THE REVIEW AREA

NWI code	Description	Acres inside
PEM1C	Palustrine emergent, persistent, seasonally flooded	4.8389
PSS1/EM1Ad	Palustrine scrub-shrub, broad-leaved deciduous / emergent, persistent, temporarily flooded, partly drained/ditched	4.2706
PFO1A	Palustrine forested, broad-leaved deciduous, temporarily flooded	3.2774
PEM1C	Palustrine emergent, persistent, seasonally flooded	0.4142
PEM1C	Palustrine emergent, persistent, seasonally flooded	0.0951
PFO1/SS1A	Palustrine forested / scrub-shrub, broad-leaved deciduous, temporarily flooded	0.0269
Total		**12.9231**

MN PUBLIC WATERS INVENTORY (PWI) NEAR THE REVIEW AREA

PWI ID	Name	Class	Distance
13005903	North Sunrise Pool	Public Water Basin	335 ft
13019a	Sunrise River	Public Ditch/Altered Natural Watercourse	1,267 ft

TRUE NORTH

0 100 200 300 400 ft

Scale: 1 inch = 300 feet (1:3,600)

when printed at 11 × 17 in (full size)

Wetlands: USFWS National Wetlands Inventory, Wetlands MapServer, queried 2026-10-09. NWI project at this site (USFWS Data_Source service): "MN Update", 2010 CIR imagery (1-ft resolution) — the MN DNR-coordinated NWI update for east-central Minnesota (project report, 2013-05-15). NWI is a photo-interpreted screening layer.
Public waters: MN DNR Public Waters Inventory basin and watercourse delineations, MnGeo service water_mn_public_waters, queried 2026-10-09 (basin delineation date 2008-01-01; watercourse record published 2016-07-28 (content date 2014-09-02)).
Imagery: USDA NAIP 2023, natural color, 0.3 m, orthorectified; acquired 2023-07-11 (quarter-quads m_4509233 NE, NW, SE, SW). Source: USDA-FSA-APFO tiles via the Microsoft Planetary Computer STAC (public). Mid-growing-season, leaf-on; no wet-season or leaf-off image was used.
Coordinates: NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database). Latitude/longitude: decimal degrees, NAD83(2011); the GeoPackage and the ORM CSV are in WGS 84 (G2139), transformed with PROJ at epoch 2026.77 (1.4 m from NAD83(2011) here).

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WETLAND DELINEATION MAP SET — MINNESOTA

Sample site · Chisago County, Minnesota · NE¼ NW¼ Sec. 14, T34N R21W

Sample site — demonstration only (SYNTHETIC field data)

USACE St. Paul District + MN BWSR · joint delineation report guidance v2.0 (2015)

FEMA FLOOD HAZARD CONTEXT (FIGURE 5)

LEGEND

Review area (sample site, 39.85 ac)

FEMA flood zones (MN DNR copy of the FEMA DFIRM database)

Zone A, special flood hazard area (no base flood elevation)

Zone X, area of minimal flood hazard (unshaded)

MN DNR PWI basin outline (for reference)

FEMA FLOOD HAZARD — CONTEXT

Item	Value
FEMA DFIRM ID	27025C (Chisago County, MN)
FIRM panels (review area)	27025C0260D and 27025C0275D, effective 2012-04-17 (MN DNR copy of the FEMA DFIRM database)
Within the review area	Zone X (area of minimal flood hazard), 39.8492 ac; no special flood hazard area (Zone A 0.0000 ac in both copies)
Flood zone mapped nearby	Zone A (1-percent-annual-chance flood hazard, approximate study, no base flood elevation)
Nearest Zone A boundary	308 ft from the review area (Esri NFHL copy: 305 ft)
Zones drawn, panels listed	Zones drawn from layer 5 and panels listed from layer 6 of the MN DNR copy of the FEMA DFIRM database, queried 2026-10-09; the panel boundary crosses the review area and is not drawn (no panel geometry was obtained)
Cross-check	FEMA NFHL as redistributed by Esri (Living Atlas), data last edited 2026-03-11; study 27025C_STUDY1, NFHL version 1.1.1.0

The FIRM panels, effective date and flood zones are taken from the MN DNR copy of the FEMA DFIRM database and are NOT confirmed against FEMA. Confirm on the effective FIRM (msc.fema.gov) before relying on them. Floodway, base flood elevations and local floodplain / shoreland ordinances were not checked.

TRUE NORTH

0 100 200 300 400 ft

Scale: 1 inch = 300 feet (1:3,600)

when printed at 11 × 17 in (full size)

Flood zones and FIRM panels: MN DNR copy of the FEMA DFIRM database (MN Geospatial Commons service water_dnr_fema_dfirm, layers 5 "Flood Hazard Zones" and 6 "Dfirm Panel Scheme", queried 2026-10-09). FEMA's final flood elevation determination for Chisago County: 77 FR 21471 (2012-04-10, docket FEMA-B-1134).
Cross-check: FEMA National Flood Hazard Layer as redistributed by Esri (Living Atlas layer "USA Flood Hazard Reduced Set"; data last edited 2026-03-11; queried 2026-10-09): the same Zone A, 305 ft from the review area.
MN DNR PWI basin outline shown for reference only (see Figure 4).
Imagery: USDA NAIP 2023, natural color, 0.3 m, orthorectified; acquired 2023-07-11 (quarter-quads m_4509233 NE, NW, SE, SW). Source: USDA-FSA-APFO tiles via the Microsoft Planetary Computer STAC (public). Mid-growing-season, leaf-on; no wet-season or leaf-off image was used.
Coordinates: NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database). Latitude/longitude: decimal degrees, NAD83(2011); the GeoPackage and the ORM CSV are in WGS 84 (G2139), transformed with PROJ at epoch 2026.77 (1.4 m from NAD83(2011) here).

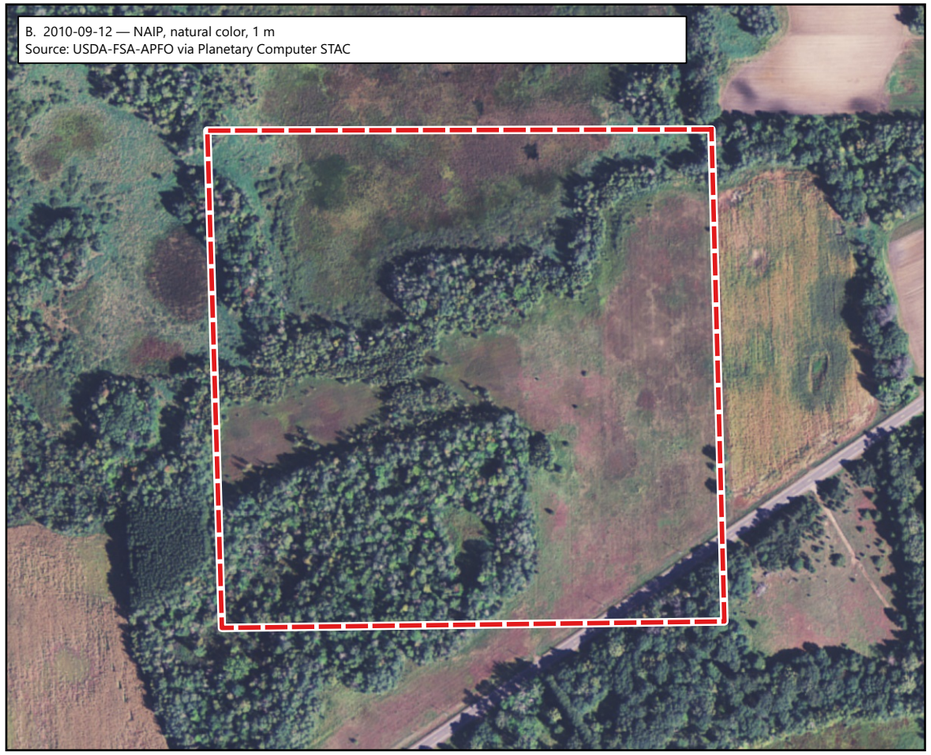
Prepared and revised by: Danir Kantserov — GIS/CAD mapping
Organization: Independent contractor (white-label sample)
Date prepared: 2026-10-09 · Rev. 3.1 (2026-10-10)
Field data: SYNTHETIC (no field survey) · Scientist of record: none
Map file: SAMPLE_MN_Delineation_MapSet.pdf

SAMPLE — not for regulatory submittal

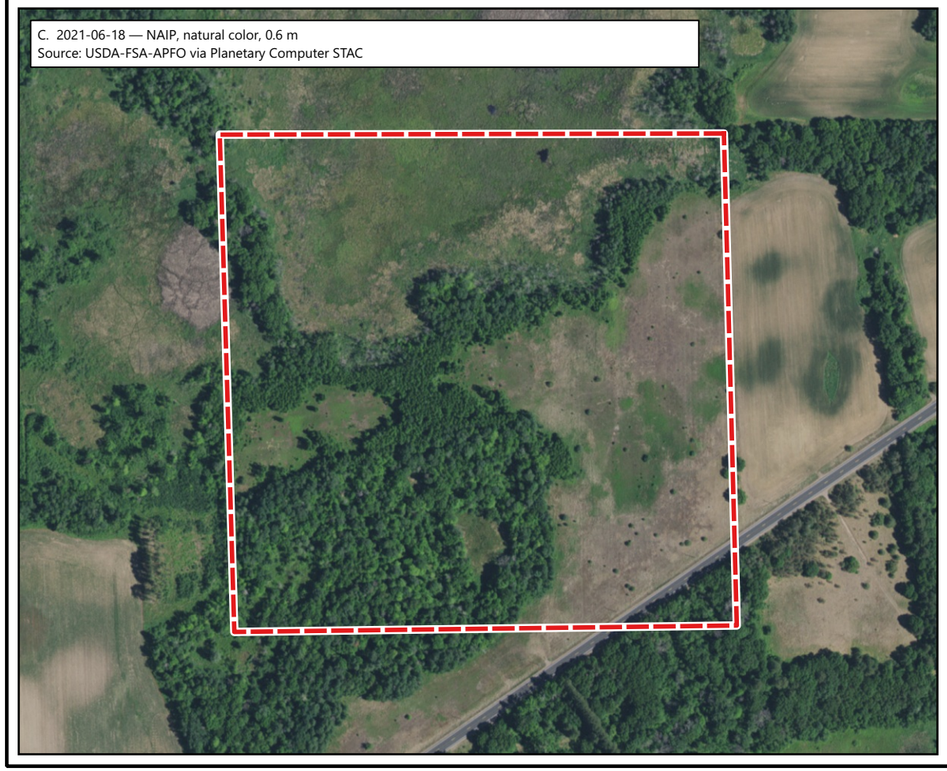
SAMPLE — SYNTHETIC FIELD DATA FOR DEMONSTRATION · NOT A WETLAND DELINEATION · NOT FOR REGULATORY SUBMITTAL



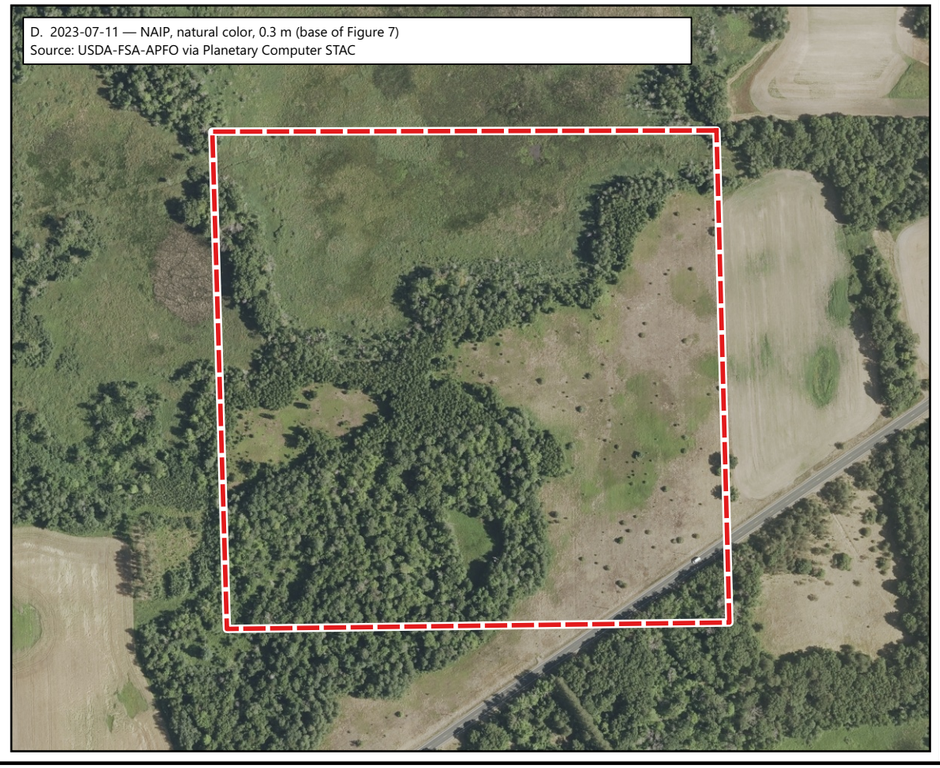
A. 1991 — USGS digital orthophoto, black and white, 1 m
Source: MnGeo WMS layer "doq" (1991 bw USGS); per-tile date not published



B. 2010-09-12 — NAIP, natural color, 1 m
Source: USDA-FSA-APFO via Planetary Computer STAC



C. 2021-06-18 — NAIP, natural color, 0.6 m
Source: USDA-FSA-APFO via Planetary Computer STAC



D. 2023-07-11 — NAIP, natural color, 0.3 m (base of Figure 7)
Source: USDA-FSA-APFO via Planetary Computer STAC

WETLAND DELINEATION MAP SET — MINNESOTA

Sample site · Chisago County, Minnesota · NE¼ NW¼ Sec. 14, T34N R21W

Sample site — demonstration only (SYNTHETIC field data)

USACE St. Paul District + MN BWSR · joint delineation report guidance v2.0 (2015)

RECENT AND HISTORICAL AERIAL REVIEW (FIGURE 6)

LEGEND

Review area (sample site, 39.85 ac)

IMAGES ON THIS SHEET

Panel	Date	Image	Pixel	Source
A	1991	USGS digital orthophoto (DOQ), black and white	1 m	MnGeo WMS layer "doq"; per-tile date not published; registered to NAIP 2023 (RMS 4.2 ft)
B	2010-09-12	NAIP, natural color	1 m	USDA-FSA-APFO via Planetary Computer STAC
C	2021-06-18	NAIP, natural color	0.6 m	USDA-FSA-APFO via Planetary Computer STAC
D	2023-07-11	NAIP, natural color (base of Figure 7)	0.3 m	USDA-FSA-APFO via Planetary Computer STAC

NAIP YEARS OVER THE SITE (PLANETARY COMPUTER STAC)

Year	Acquired	Pixel	Tiles (quarter quads)	On this sheet
2010	2010-09-12	1 m	4509233 NE NW SE SW	panel B
2013	2013-07-03, 2013-08-17	1 m	4509233 NE NW SE SW	listed only
2015	2015-09-21	1 m	4509233 NE NW SE SW	listed only
2017	2017-07-28, 2017-09-21	1 m	4509233 NE NW SE SW	listed only
2019	2019-07-31	0.6 m	4509233 NE NW SE SW	listed only
2021	2021-06-18	0.6 m	4509233 NE NW SE SW	panel C
2023	2023-07-11	0.3 m	4509233 NE NW SE SW	panel D and Figure 7

TRUE NORTH

0 100 200 300 400 ft

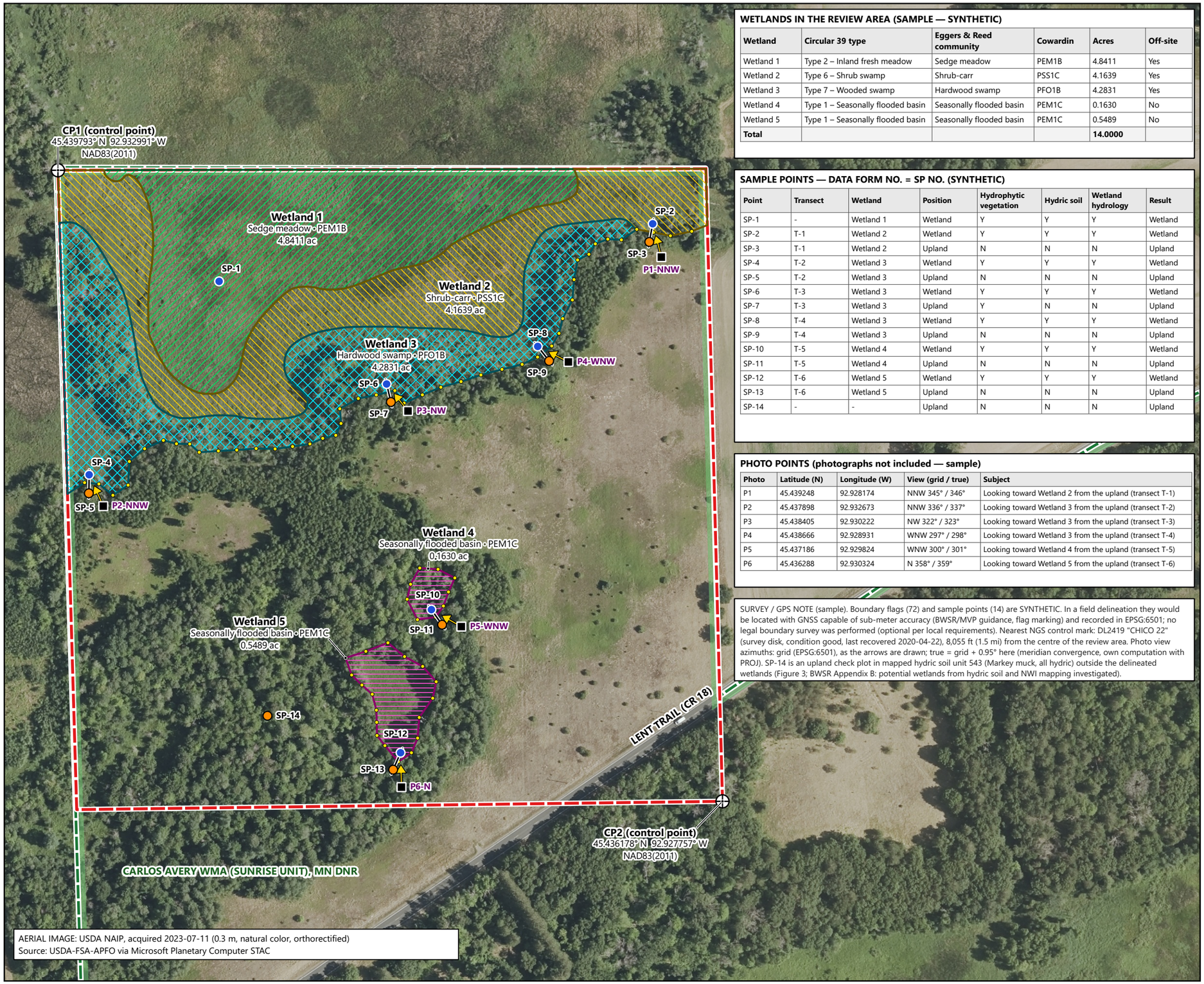
Scale: 1 inch = 400 feet (1:4,800)
when printed at 11 × 17 in (full size)

Four images of the same area at one scale (1 inch = 400 feet). Recent: NAIP 2023-07-11. Historical: 1991, 2010-09-12, 2021-06-18.
NAIP images are summer (leaf-on). Antecedent precipitation / normal-conditions screening for each image date (BWSR/MVP guidance, section 3.7) was NOT evaluated in this sample.
Panel A (1991 DOQ): placed with the extent of the WMS request that returned the image and registered to NAIP 2023 along Iris Avenue and Lent Trail: 17 road stations, residual RMS 4.2 ft, largest 10.5 ft (shift applied 1.7 ft; OpenStreetMap centrelines).
A 2025 NAIP mosaic is published by MnGeo but no per-tile acquisition date could be obtained, so it is not used.
Coordinates: NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database). Latitude/longitude: decimal degrees, NAD83(2011); the GeoPackage and the ORM CSV are in WGS 84 (G2139), transformed with PROJ at epoch 2026.77 (1.4 m from NAD83(2011) here).

Prepared and revised by: Danir Kantserov — GIS/CAD mapping
Organization: Independent contractor (white-label sample)
Date prepared: 2026-10-09 · Rev. 3.1 (2026-10-10)
Field data: SYNTHETIC (no field survey) · Scientist of record: none
Map file: SAMPLE_MN_Delineation_MapSet.pdf

SAMPLE — not for regulatory submittal

SAMPLE — SYNTHETIC FIELD DATA FOR DEMONSTRATION · NOT A WETLAND DELINEATION · NOT FOR REGULATORY SUBMITTAL



AERIAL IMAGE: USDA NAIP, acquired 2023-07-11 (0.3 m, natural color, orthorectified)
Source: USDA-FSA-APFO via Microsoft Planetary Computer STAC

WETLANDS IN THE REVIEW AREA (SAMPLE — SYNTHETIC)

Wetland	Circular 39 type	Eggers & Reed community	Cowardin	Acres	Off-site
Wetland 1	Type 2 – Inland fresh meadow	Sedge meadow	PEM1B	4.8411	Yes
Wetland 2	Type 6 – Shrub swamp	Shrub-carr	PSS1C	4.1639	Yes
Wetland 3	Type 7 – Wooded swamp	Hardwood swamp	PFO1B	4.2831	Yes
Wetland 4	Type 1 – Seasonally flooded basin	Seasonally flooded basin	PEM1C	0.1630	No
Wetland 5	Type 1 – Seasonally flooded basin	Seasonally flooded basin	PEM1C	0.5489	No
Total				**14.0000**	

SAMPLE POINTS — DATA FORM NO. = SP NO. (SYNTHETIC)

Point	Transect	Wetland	Position	Hydrophytic vegetation	Hydric soil	Wetland hydrology	Result
SP-1	-	Wetland 1	Wetland	Y	Y	Y	Wetland
SP-2	T-1	Wetland 2	Wetland	Y	Y	Y	Wetland
SP-3	T-1	Wetland 2	Upland	N	N	N	Upland
SP-4	T-2	Wetland 3	Wetland	Y	Y	Y	Wetland
SP-5	T-2	Wetland 3	Upland	N	N	N	Upland
SP-6	T-3	Wetland 3	Wetland	Y	Y	Y	Wetland
SP-7	T-3	Wetland 3	Upland	Y	N	N	Upland
SP-8	T-4	Wetland 3	Wetland	Y	Y	Y	Wetland
SP-9	T-4	Wetland 3	Upland	N	N	N	Upland
SP-10	T-5	Wetland 4	Wetland	Y	Y	Y	Wetland
SP-11	T-5	Wetland 4	Upland	N	N	N	Upland
SP-12	T-6	Wetland 5	Wetland	Y	Y	Y	Wetland
SP-13	T-6	Wetland 5	Upland	N	N	N	Upland
SP-14	-	-	Upland	N	N	N	Upland

PHOTO POINTS (photographs not included — sample)

Photo	Latitude (N)	Longitude (W)	View (grid / true)	Subject
P1	45.439248	92.928174	NNW 345° / 346°	Looking toward Wetland 2 from the upland (transect T-1)
P2	45.437898	92.932673	NNW 336° / 337°	Looking toward Wetland 3 from the upland (transect T-2)
P3	45.438405	92.930222	NW 322° / 323°	Looking toward Wetland 3 from the upland (transect T-3)
P4	45.438666	92.928931	WNW 297° / 298°	Looking toward Wetland 3 from the upland (transect T-4)
P5	45.437186	92.929824	WNW 300° / 301°	Looking toward Wetland 4 from the upland (transect T-5)
P6	45.436288	92.930324	N 358° / 359°	Looking toward Wetland 5 from the upland (transect T-6)

SURVEY / GPS NOTE (sample). Boundary flags (72) and sample points (14) are SYNTHETIC. In a field delineation they would be located with GNSS capable of sub-meter accuracy (BWSR/MVP guidance, flag marking) and recorded in EPSG:6501; no legal boundary survey was performed (optional per local requirements). Nearest NGS control mark: DL2419 "CHICO 22" (survey disk, condition good, last recovered 2020-04-22), 8,055 ft (1.5 mi) from the centre of the review area. Photo view azimuths: grid (EPSG:6501), as the arrows are drawn; true = grid + 0.95° here (meridian convergence, own computation with PROJ). SP-14 is an upland check plot in mapped hydric soil unit 543 (Markey muck, all hydric) outside the delineated wetlands (Figure 3; BWSR Appendix B: potential wetlands from hydric soil and NWI mapping investigated).

WETLAND DELINEATION MAP SET — MINNESOTA

Sample site · Chisago County, Minnesota · NE¼ NW¼ Sec. 14, T34N R21W

Sample site — demonstration only (SYNTHETIC field data)

USACE St. Paul District + MN BWSR · joint delineation report guidance v2.0 (2015)

FINAL DELINEATION ON NAIP 2023 (FIGURE 7)

LEGEND

- Review area (sample site, 39.85 ac)
- Carlos Avery WMA boundary (Sunrise Unit), MN DNR
- Wetland 1 — Sedge meadow, PEM1B (4.8411 ac)
- Wetland 2 — Shrub-carr, PSS1C (4.1639 ac)
- Wetland 3 — Hardwood swamp, PFO1B (4.2831 ac)
- Wetland 4 — Seasonally flooded basin, PEM1C (0.1630 ac)
- Wetland 5 — Seasonally flooded basin, PEM1C (0.5489 ac)
- Wetland boundary flag (synthetic; 72 flags, IDs in the GIS/DXF files)
- Sample point, wetland plot (label = data form no.)
- Sample point, upland plot (label = data form no.)
- Transect T-n through a paired wetland/upland plot
- Photo point (P#-direction) and view direction
- Control point (latitude/longitude, NAD83(2011))

DELINEATION SUMMARY

Wetlands in the review area: 5, 14.0000 ac (review area 39.85 ac)
Wetlands 1–3 (one complex): 13.2881 ac
Wetlands 4–5 (depressional basins): 0.7119 ac
Sample points: 14, data forms 1–14 (6 transects; SP-1, SP-14 single plots)
Boundary flags: 72 (nearest-flag spacing: median 33 ft, 93 % within 43 ft, max 45 ft)
Photo points: 6 (P1–P6), locations referenced

TRUE NORTH

0 100 200 300 400 ft

Scale: 1 inch = 200 feet (1:2,400)

when printed at 11 × 17 in (full size)

Land status: 39.83 of the 39.85 ac review area lie inside the Carlos Avery Wildlife Management Area boundary (Sunrise Unit, MN DNR; layer bdry_dnr_wildlife_mgmt_areas_pub, MN Geospatial Commons, edited 2024-01-25, queried 2026-10-09); PAD-US 4.1 records MN DNR fee ownership on 36.17 ac (2015 boundary). The field data are synthetic; no finding is made about the property.
Imagery: USDA NAIP 2023, natural color, 0.3 m, orthorectified; acquired 2023-07-11 (quarter-quads m_4509233 NE, NW, SE, SW). Source: USDA-FSA-APFO tiles via the Microsoft Planetary Computer STAC (public). Mid-growing-season, leaf-on; no wet-season or leaf-off image was used.
Coordinates: NAD83(2011) / Minnesota Central, US survey feet (EPSG:6501, verified in the QGIS CRS database). Latitude/longitude: decimal degrees, NAD83(2011); the GeoPackage and the ORM CSV are in WGS 84 (G2139), transformed with PROJ at epoch 2026.77 (1.4 m from NAD83(2011) here).
Wetland boundaries, flags, sample points, transects and photo points are SYNTHETIC (derived from NWI, SSURGO hydric soils and the lidar DEM for demonstration). No field delineation was performed. Jurisdictional status is not indicated.
Wetland types: Circular 39 / Eggers & Reed / Cowardin per BWSR crosswalk (2025)
Wetlands 1–3 are contiguous community units of one wetland complex; thin lines inside it are community boundaries, not wetland–upland boundaries (no flags). Areas: planar, from the GIS geometry (EPSG:6501); acres rounded once (4 decimals for wetlands); totals are sums of the rounded rows shown.

Prepared and revised by: Danir Kantserov — GIS/CAD mapping
Organization: Independent contractor (white-label sample)
Date prepared: 2026-10-09 · Rev. 3.1 (2026-10-10)
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